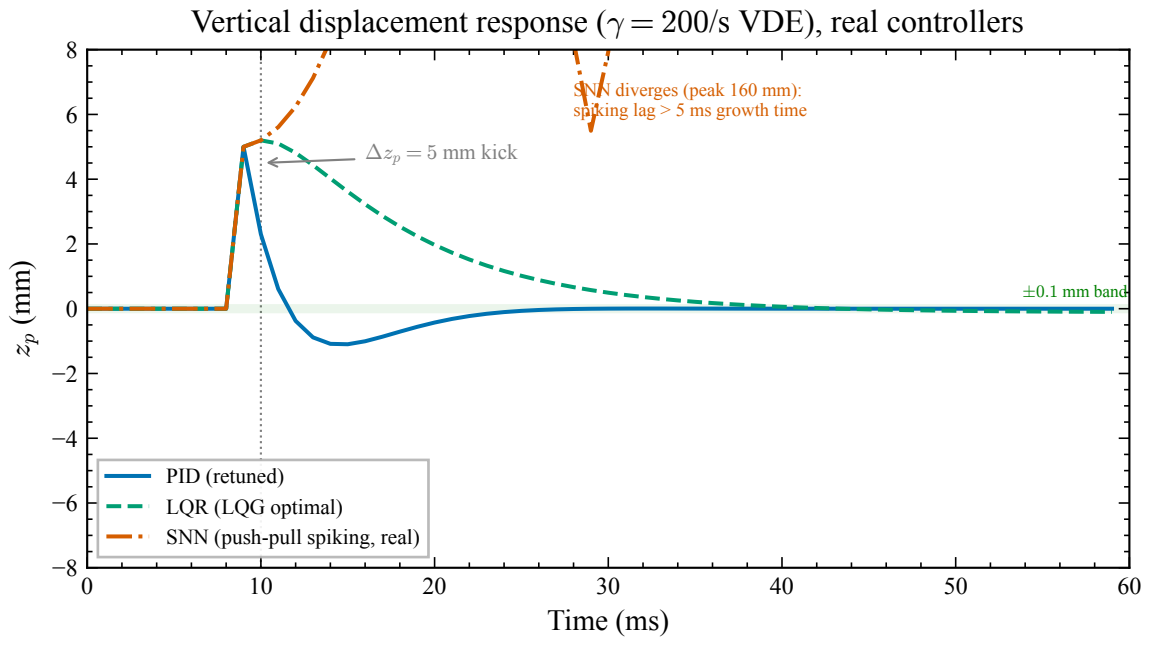
Vertical displacement response ($\gamma = 200$/s VDE), real controllers
SNN diverges (peak 160 mm):
spiking lag > 5 ms growth time
$\Delta z_p = 5$ mm kick
±0.1 mm band
PID (retuned)
LQR (LQG optimal)
SNN (push-pull spiking, real)
z_p (mm)
Time (ms)
8
6
4
2
0
−2
−4
−6
−8
0
10
20
30
40
50
60